BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Utility Software

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why does an operating system use a device driver?

2. What does anti-malware detect?
   A) Every future threat
   B) All hardware faults
   C) Malicious software
   D) All weak passwords

3. Why use different user access levels?

4. A character code uses exactly 7 bits. How many different characters can it represent? Can it encode 129 different characters without more bits?

Today's objectives

Identify the purposes of three OCR J277 utilities. Explain how defragmentation applies to magnetic hard disks.

Key vocabulary - Read, cover, recall

Utility software
Programs performing additional maintenance or protection tasks.

Encryption
Converts plaintext into ciphertext using an algorithm and key.

Decryption
Converts encrypted data back to readable form with the required key.

Compression
Reduces how much storage data requires.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Utilities versus the OS

Utility software performs additional housekeeping, maintenance and protection tasks which might not be performed directly by the operating system.

Encryption utility

Encryption changes readable plaintext into ciphertext using an algorithm/key, protecting data if it is accessed without authorisation. A suitable key is needed to decrypt it.

Compression utility

Compression reduces the storage space needed for data. Lossless allows exact reconstruction; lossy removes information and is unsuitable when exact recovery is needed.

Defragmentation

On a magnetic hard disk, fragments of a file may be scattered. Defragmentation puts file parts together so the read head needs fewer movements. SSDs should not be routinely defragmented for this purpose.

I Do - Worked model

A school library's magnetic hard disk has become fragmented. Explain how defragmentation can improve file access. [4 marks]

- Identify the storage medium: a magnetic hard disk.
- Parts of files may be scattered in separate locations on disk [1].
- Defragmentation rearranges these parts into adjacent blocks [1], reducing movement of the disk read/write head [1].
- Less movement can reduce time accessing the file [1]. Do not suggest this improves SSD read-head movement.

We Do - Guided practice

1. A business stores confidential customer records on portable media. Explain why encryption software is suitable. [4 marks]

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2. A podcaster wants to reduce the size of large audio files and accepts some lost quality. Explain a suitable utility and one drawback. [4 marks]

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3. A school has an SSD and a magnetic hard disk. Explain why defragmentation may help the HDD but is not normally useful for the SSD. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the line that prints storage saved in KB.

```
originalKB = 240
compressedKB = 60
```

You Do - Independent exam questions

Q2. Name the utility that rearranges file blocks on a magnetic hard disk. [1 mark]

Q3. A doctor encrypts confidential files on a portable drive. Explain TWO reasons this helps keep information private. [4 marks]

Q4. A school is choosing utility software. Explain when compression, encryption and defragmentation may each be useful. [6 marks]

Exit Ticket - True or False?

Defragmentation is mainly intended to reduce mechanical read-head movement on fragmented magnetic hard disks.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It enables communication with and control of a particular peripheral.

2. C - Malicious software

3. Give users only the access rights they need. This limits access without permission and possible damage.

4. (a) 2^7 = 128; no, one more character requires more distinct codes.

I Do - Explain the model

Identify the storage medium: a magnetic hard disk.

Parts of files may be scattered in separate locations on disk [1].

Defragmentation rearranges these parts into adjacent blocks [1], reducing movement of the disk read/write head [1].

Less movement can reduce time accessing the file [1]. Do not suggest this improves SSD read-head movement.

We Do - Model answers

1. Encryption turns readable data into ciphertext [1] using an algorithm/key [1]. If the device is lost or stolen [1], the records cannot readily be understood without the correct decryption key [1].

2. Compression software reduces file size [1], saving storage or transfer time [1]. Lossy compression discards some data [1], so the original sound cannot be perfectly reconstructed [1].

3. On an HDD files can be fragmented [1], making the mechanical read/write head move between locations [1]. Defragmentation groups parts and reduces head movement [1]; SSDs have no moving read/write head, so this benefit does not apply [1].

Code First - Answer

```
originalKB = 240
compressedKB = 60
print(originalKB - compressedKB)
```

Expected output: 180

You Do - Self-mark

Q2: Defragmentation [1].

Q3: Encryption transforms readable data into ciphertext [1], preventing straightforward reading [1]. A decryption key is needed [1], so losing the drive does not automatically expose readable records [1].

Q4: Compression reduces file size [1] to save storage or transfer time [1]. Encryption converts data to unreadable ciphertext [1], protecting content from unauthorised access [1]. Defragmentation reorganises fragmented HDD files [1] and can reduce mechanical disk access time [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.